APEX FX TRADING BOT

Product Requirements Document

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Document Owner	Product & Engineering Team
Instruments	EUR/USD, GBP/USD, USD/JPY, USD/CHF, USD/CAD, XAU/USD (Gold)
Target Platform	MetaTrader 5 (MT5) via Python MT5 API
Classification	**CONFIDENTIAL — Internal Use Only**

1. EXECUTIVE SUMMARY

APEX FX Trading Bot is a fully automated, multi-instrument algorithmic trading system engineered to trade the five major forex currency pairs (EUR/USD, GBP/USD, USD/JPY, USD/CHF, USD/CAD) alongside the gold commodity pair XAU/USD. The bot runs 24 hours a day, five days a week, executing trades through MetaTrader 5 via a Python-based execution engine, with each instrument governed by a bespoke strategy and an instrument-specific risk management framework.

Unlike generic trading bots that apply a one-size-fits-all strategy, APEX FX is built on the foundational insight that each instrument has a distinct volatility profile, liquidity depth, session sensitivity, and macroeconomic driver. The system therefore assigns dedicated strategy modules to each pair and calibrates risk parameters — including position sizing, stop-loss distances, maximum daily drawdown, and risk-per-trade percentages — individually for each instrument.

Core Objectives

1. Consistent profitability across all six instruments through regime-aware strategy selection.2. Capital preservation as the primary mandate, with profitability as a secondary objective.3. Adaptive risk management: each pair's exposure is governed by its own volatility, liquidity, and correlation profile.4. Full automation: zero manual intervention required during normal operation.5. Institutional-grade execution: low-latency order routing, spread filtering, and session-based gating.

2. PRODUCT VISION & GOALS

2.1 Problem Statement

Retail and semi-institutional traders attempting to trade multiple forex pairs and commodities simultaneously face three core challenges:

- Emotional bias and lack of discipline in trade execution, especially during drawdowns.
- Inability to monitor 6 instruments across multiple timeframes in real time.
- Applying identical risk rules to fundamentally different instruments — treating gold's $30/day average range the same as EUR/USD's 80-pip range is a principal cause of over-exposure and account blowup.

2.2 Vision Statement

To build the most disciplined, risk-aware multi-instrument trading bot in the retail algorithmic trading space — one that trades like a quantitative fund manager: instrument-aware, regime-conscious, and capital-preserving first.

2.3 Success Metrics

Metric	Minimum Target	Stretch Target
Monthly Net Return	3% – 5%	7% – 10%
Maximum Drawdown (Account)	< 15%	< 10%
Win Rate (All Pairs)	> 50%	> 60%
Profit Factor	> 1.4	> 1.8
Average Risk/Reward Ratio	1:1.5	1:2.5
Sharpe Ratio (Annualized)	> 1.0	> 1.5
Max Consecutive Losses	< 8 trades	< 5 trades
System Uptime	> 99%	> 99.9%

3. INSTRUMENT PROFILES & MARKET CHARACTERISTICS

Each of the six instruments traded by APEX FX has a distinct market personality. The bot's strategy and risk engine must reflect these differences. Below is the full instrument profile for each pair.

3.1 EUR/USD — Euro vs. US Dollar

The most liquid forex pair in the world, accounting for approximately 28% of all daily forex volume ($7.5 trillion market). EUR/USD is highly sensitive to ECB and Fed monetary policy divergence, inflation data (CPI), and GDP releases from both the Eurozone and the US.

- Average Daily Range (ADR): 60 – 90 pips
- Primary Trading Sessions: London (08:00–12:00 UTC) and New York overlap (13:00–17:00 UTC)
- Volatility Profile: Low-to-moderate. Tight spreads (0.0–1.0 pips on ECN accounts)
- Key Drivers: ECB/Fed rate decisions, NFP, CPI, PMI, Eurozone trade balance
- Correlation: Strongly negatively correlated with USD Index (DXY)
- Best Strategy Fit: Trend-following (EMA crossovers) and breakout on key levels

3.2 GBP/USD — British Pound vs. US Dollar

Known as 'Cable,' GBP/USD is the third most traded forex pair globally. It is more volatile than EUR/USD due to Brexit-era structural uncertainty, Bank of England policy unpredictability, and UK political risk. It produces larger intraday swings and is susceptible to overnight gap risk.

- Average Daily Range (ADR): 80 – 130 pips
- Primary Trading Sessions: London open (07:00–12:00 UTC) is the dominant session
- Volatility Profile: Moderate-to-high. Spreads widen significantly during off-hours
- Key Drivers: BoE rate decisions, UK CPI, UK employment data, PMI, political events
- Correlation: Partially correlated with EUR/USD; diverges during UK-specific events
- Best Strategy Fit: Breakout and momentum strategies; requires wider stops

3.3 USD/JPY — US Dollar vs. Japanese Yen

The second most traded pair globally and a key barometer of risk sentiment. USD/JPY tends to trend strongly during risk-on environments (USD up, JPY weakens as carry trade is funded) and reverses sharply during risk-off events. The Bank of Japan's YCC (Yield Curve Control) policy creates unique gap risk.

- Average Daily Range (ADR): 50 – 80 pips
- Primary Trading Sessions: Tokyo (00:00–03:00 UTC) and London/New York overlap
- Volatility Profile: Moderate during normal conditions; can spike violently on BoJ intervention

- Key Drivers: Fed/BoJ rate differential, US Treasury yields, risk sentiment, BoJ intervention risk
- Correlation: Strong positive correlation with US 10-year Treasury yields
- Best Strategy Fit: Carry trade momentum, trend-following; add intervention risk filter

3.4 USD/CHF — US Dollar vs. Swiss Franc

USD/CHF is closely correlated with EUR/USD (approximately -0.85 inverse correlation) because the Swiss National Bank (SNB) pegs CHF loosely to EUR. The CHF is the world's primary safe-haven currency, and USD/CHF often moves inversely to risk sentiment. The SNB is known for surprise interventions.

- Average Daily Range (ADR): 50 – 75 pips
- Primary Trading Sessions: European and New York sessions
- Volatility Profile: Low-to-moderate; spikes during SNB communications
- Key Drivers: SNB rate decisions, risk sentiment, EUR/USD movement, geopolitical events
- Correlation: ~-0.85 with EUR/USD; use as diversification hedge within the portfolio
- Best Strategy Fit: Mean reversion during ranging conditions; trend on EUR/USD confirmation

3.5 USD/CAD — US Dollar vs. Canadian Dollar

The 'Loonie' is a commodity-linked currency pair, heavily influenced by crude oil prices (Canada is a top oil exporter to the US). USD/CAD tends to move inversely to oil prices — when oil rises, CAD strengthens and USD/CAD falls. Economic data from Canada is often overshadowed by US data and oil movements.

- Average Daily Range (ADR): 60 – 90 pips
- Primary Trading Sessions: New York session (13:00–21:00 UTC) is dominant
- Volatility Profile: Moderate; spikes on oil inventory data (EIA Wednesday), OPEC meetings, BoC decisions
- Key Drivers: WTI crude oil price, BoC rate decisions, Canadian CPI and employment, US data
- Correlation: Strong inverse correlation with crude oil prices (-0.7 to -0.9)
- Best Strategy Fit: Commodity-correlated trend following; oil as a leading indicator

3.6 XAU/USD — Gold vs. US Dollar

Gold is not a currency pair but is traded as a spot commodity denominated in USD. It is by far the most volatile instrument in this portfolio. Gold's price is driven by real interest rates (negative correlation), USD strength, geopolitical risk, central bank buying, and inflation expectations. It moves $15–$40 per ounce daily on average, equivalent to 150–400 pips — dwarfing all five forex pairs.

- Average Daily Range (ADR): $15 – $40+ per troy ounce (150–400 pip equivalent)
- Primary Trading Sessions: London open and New York session; very active overnight (Asia)

- Volatility Profile: HIGH. Has extreme volatility spikes around US CPI, FOMC, and geopolitical events
- Key Drivers: US real yields, USD Index, inflation, central bank demand, geopolitical risk, risk-off flows
- Correlation: Strong negative correlation with USD and real yields; positive with risk-off sentiment
- Best Strategy Fit: Multi-timeframe trend following on H1/H4; breakout on session opens; mean reversion
- Special Risk Consideration: Requires REDUCED position size (0.5%–1% risk vs. 1%–2% for forex pairs) due to extreme pip value and volatility

4. STRATEGY ARCHITECTURE

APEX FX employs a multi-strategy architecture. Rather than a single strategy applied uniformly, each instrument is assigned a primary and secondary strategy module. The Regime Detection Engine (RDE) determines which strategy is active based on real-time market conditions. This addresses the core problem of strategies that work in trending markets failing in ranging markets, and vice versa.

4.1 Regime Detection Engine (RDE)

The RDE runs on every new bar close and classifies the current market into one of three regimes for each instrument independently:

Regime	Detection Criteria	Active Strategy
TRENDING	ADX > 25, price above/below 50 EMA, ATR expanding	Trend-Following Module
RANGING	ADX < 20, price oscillating within Bollinger Bands, ATR contracting	Mean Reversion Module
BREAKOUT PENDING	Bollinger Band Squeeze (width < 20-period average), pre-major event	Breakout Module
AVOID / NO TRADE	ADX between 20–25 (ambiguous), major news within 30 min	No new trades opened

4.2 EUR/USD Strategy: Multi-Timeframe Trend Following + Session Breakout

Primary Strategy: EMA Crossover Trend System

Given EUR/USD's high liquidity and reliable trend behavior during London-NY overlap, the primary strategy is a multi-timeframe EMA crossover system with confirmation filters.

- Timeframes: H4 for trend direction, H1 for entry timing
- Indicators: 20 EMA, 50 EMA, 200 EMA (H4); RSI(14), MACD(12,26,9) for confirmation
- Entry Signal: Fast EMA (20) crosses above/below Slow EMA (50) on H1, confirmed by H4 trend direction
- Entry Filter: RSI must not be in overbought (>70) zone for longs, or oversold (<30) for shorts
- Volume Filter: Spread must be < 1.5 pips at entry time
- Session Gate: Only active during London open (07:00–09:00 UTC) and London-NY overlap (13:00–16:00 UTC)

Secondary Strategy: London Session Breakout

- Identify the high/low range formed from 06:00–08:00 UTC (pre-London accumulation)
- Enter on breakout above/below this range with a candle close confirmation on M15
- This strategy is activated only when RDE classifies the regime as BREAKOUT PENDING

4.3 GBP/USD Strategy: Momentum Breakout + Fibonacci Retracement

Primary Strategy: London Open Momentum

GBP/USD produces its most reliable moves in the first two hours of the London session. The strategy capitalizes on institutional order flow during this window.

- Timeframes: H1 for trend, M15 for entry
- Indicators: 50 EMA (H1), RSI(14), ATR(14), Bollinger Bands(20,2)
- Entry Signal: Price breaks above/below Bollinger Band outer band on M15 with ATR above 30-period average
- Confirmation: Price must be on the correct side of H1 50 EMA
- Session Gate: 07:00–11:00 UTC only. No new trades after 14:00 UTC (avoids daily settlement spread widening)
- Spread Filter: Entry blocked if spread > 2.0 pips

Secondary Strategy: Fibonacci Pullback

- In trending conditions (ADX > 30), enter on 50% or 61.8% Fibonacci retracements of the last confirmed swing
- Minimum swing size: 60 pips. Must be confirmed by RSI divergence or candle pattern (bullish/bearish engulfing)

4.4 USD/JPY Strategy: Carry Trade Momentum + Trend Continuation

Primary Strategy: Multi-Session Trend with BoJ Intervention Filter

USD/JPY is a trending pair by nature, driven by the US-Japan interest rate differential. The strategy rides this momentum while incorporating a hard stop for BoJ intervention risk.

- Timeframes: D1 for macro bias, H4 for trend, H1 for entry
- Indicators: 200 EMA (D1), 50 EMA (H4), Stochastic(5,3,3), ATR(14)
- Entry Signal: Price above 200 EMA (bullish bias), 50 EMA sloping upward, Stochastic crosses above 20 in oversold territory
- Intervention Filter: If price moves > 200 pips in a single H4 candle in the same direction as the trade, pause new entries for 24 hours — this pattern is consistent with BoJ intervention activity
- Session Gate: Avoid trading during Tokyo open news windows (23:50–00:30 UTC) when BoJ or Japanese economic data releases are scheduled

4.5 USD/CHF Strategy: Mean Reversion + EUR/USD Divergence

Primary Strategy: Bollinger Band Mean Reversion

Because USD/CHF is strongly inversely correlated to EUR/USD, it tends to be range-bound more often than trending. Mean reversion is statistically superior for this pair in the absence of a strong EUR/USD trend.

- Timeframes: H1 for signals, H4 for confirmation
- Indicators: Bollinger Bands(20,2), RSI(14), Williams %R(14)
- Entry Signal: Price touches outer Bollinger Band, RSI is overbought/oversold, Williams %R confirms
- EUR/USD Divergence Filter: Only trade USD/CHF mean reversion when EUR/USD is also ranging (ADX < 20). If EUR/USD is trending strongly, USD/CHF will trend in the opposite direction — switch to trend mode.
- Session Gate: European session (07:00–15:00 UTC) only; avoid NY close session where CHF liquidity drops

4.6 USD/CAD Strategy: Oil-Correlated Trend Following

Primary Strategy: WTI Crude Oil Correlation Trend

USD/CAD's strong inverse relationship with oil prices provides a powerful leading indicator that no purely technical system can replicate. The strategy uses oil as an external signal to confirm or filter technical setups.

- Timeframes: H4 for oil and pair trend, H1 for entry
- Indicators: 50 EMA, 200 EMA, MACD(12,26,9), WTI crude oil price feed (secondary data)
- Entry Signal: EMA crossover on H1 CONFIRMED by WTI crude trending in the opposite direction (oil up = USD/CAD down = sell signal; oil down = USD/CAD up = buy signal)
- Oil EIA Filter: On Wednesday 15:30 UTC (EIA inventory release), no new trades open within 30 minutes before or after
- Session Gate: New York session dominance (13:00–20:00 UTC)
- OPEC Event Filter: During scheduled OPEC+ meetings, reduce position size by 50% due to binary outcome risk

4.7 XAU/USD Strategy: Multi-Timeframe Trend + Breakout Hybrid

Primary Strategy: 200 EMA Trend System with ATR Stops

Gold's dominant characteristic is strong multi-week trends driven by macroeconomic themes. The primary strategy is a trend-following system on the H4/Daily timeframe, exploiting these sustained directional moves.

- Timeframes: D1 for macro trend direction, H4 for entry context, H1 for precise entry
- Indicators: 200 EMA (D1), 50 EMA (H4), RSI(14), ATR(14), MACD confirmation

- Entry Signal: Price pulls back to 50 EMA on H4 while above 200 EMA on D1 (bullish trend); RSI between 40–60 on H4 (neutral, not exhausted); MACD histogram flattening or turning positive
- Gold-Specific News Filter: Hard block on new entries within 45 minutes of: US CPI, FOMC decisions, NFP, US GDP, geopolitical flashpoints. Gold's reaction to these events is too unpredictable for entry — but can be traded as a breakout 15 minutes AFTER the release
- Session Focus: London open and New York open are peak gold volatility windows

Secondary Strategy: Session Open Breakout

- At London open (08:00 UTC) and New York open (13:00 UTC), calculate the 30-minute opening range (OR)
- Enter on confirmed break above/below the OR high/low on M15 close
- Minimum breakout size: $5 (50 pips). Avoid false breakouts by requiring volume confirmation
- This strategy is particularly effective during macro news days when gold gaps and moves strongly

5. RISK MANAGEMENT FRAMEWORK

Risk management is the most critical component of APEX FX. Every instrument receives a tailored risk profile. The overarching philosophy is: the more volatile the instrument, the smaller the position size and the wider (but proportionally calculated) the stop-loss. No fixed stop-loss sizes are used; all stops are ATR-based to adapt to current market conditions.

5.1 Core Risk Principles

- Capital preservation is always the primary objective. Profitability is secondary.
- All stop-losses are mandatory and hard (not trailing-only). No trade is ever opened without a stop-loss.
- Position sizing is calculated dynamically per trade based on account balance, instrument volatility (ATR), and the specific risk percentage allocated to that instrument.
- Correlated positions are monitored. EUR/USD and USD/CHF cannot both be at maximum position size simultaneously due to their inverse correlation. GBP/USD and EUR/USD positions are capped jointly.
- A global account drawdown circuit breaker halts ALL trading if the account loses more than 8% in a single day or 15% in a rolling 30-day window.

5.2 Position Sizing Formula

Position size (lots) for every trade is calculated as follows:

Position Size Formula

Lot Size = (Account Balance × Risk % per Trade) ÷ (ATR Multiplier × ATR(14) × Pip Value)Where: - Account Balance = Current equity (not initial balance) - Risk % = Instrument-specific (see Section 5.3) - ATR Multiplier = 1.5× for forex pairs, 2.0× for XAU/USD - Pip Value = Broker-specific tick value (fetched dynamically from MT5 API, never hardcoded) - Minimum lot: 0.01 | Maximum lot per trade: 5.0 (overridden by position cap)

5.3 Per-Instrument Risk Parameters

Pair	Risk/Trade	ATR Mult.	Max Daily Loss	Max DD	Max Lot	Max Pos.
EUR/USD	1.5%	1.5× ATR	3%	8%	2.0	2
GBP/USD	1.2%	1.8× ATR	3%	8%	1.5	2
USD/JPY	1.5%	1.5× ATR	3%	8%	2.0	2
USD/CHF	1.0%	1.5× ATR	2.5%	7%	1.5	1

Pair	Risk/Trade	ATR Mult.	Max Daily Loss	Max DD	Max Lot	Max Pos.
USD/CAD	1.2%	1.5× ATR	3%	8%	1.5	2
XAU/USD	0.75%	2.0× ATR	2%	6%	0.5	1

Rationale for XAU/USD reduced risk: Gold's ATR is typically $15–$40 per ounce. At 0.01 lots (1 oz), a 200-pip ($2.00) move equals $2. But at 0.5 lots (50 oz), a $20 ATR move equals $1,000. The 0.75% risk cap and 2.0× ATR multiplier ensure the stop-loss is placed beyond daily noise, while the reduced risk percentage protects the account from gold's outsized moves.

5.4 Stop-Loss & Take-Profit Strategy

Pair	Stop-Loss Method	Take-Profit Method	Trailing Stop
EUR/USD	1.5× ATR(14) below/above entry	2.5× ATR (RR = 1:1.67)	Activated at 1.0× ATR profit; trails by 1.0× ATR
GBP/USD	1.8× ATR(14)	3.0× ATR (RR = 1:1.67)	Activated at 1.5× ATR profit; trails by 1.2× ATR
USD/JPY	1.5× ATR(14) + 10 pips buffer for intervention risk	2.5× ATR	Activated at 1.0× ATR profit; trails by 1.0× ATR
USD/CHF	1.5× ATR(14)	2.0× ATR (RR = 1:1.33) — tighter due to range-bound nature	Activated at 0.8× ATR profit; trails by 0.8× ATR
USD/CAD	1.5× ATR(14)	2.5× ATR	Activated at 1.0× ATR profit; trails by 1.0× ATR
XAU/USD	2.0× ATR(14) — MANDATORY wider stop for gold noise	3.5× ATR (RR = 1:1.75)	Activated at 1.5× ATR profit; trails by 1.5× ATR — gold trends are stronger and longer

5.5 Portfolio-Level Risk Controls

Daily Circuit Breakers

- Single pair daily loss exceeds its per-pair daily limit: That pair is locked for the remainder of the trading day.
- Total account drawdown exceeds 8% in a single day: ALL pairs are locked. Bot sends alert and awaits manual review.
- Total account drawdown exceeds 15% in a rolling 30-day window: ALL pairs are locked. Bot enters defensive mode.

Correlation Controls

- EUR/USD + USD/CHF: Cannot hold positions in the same direction simultaneously (they are inversely correlated — a bullish EUR/USD position and a bullish USD/CHF position are effectively hedging each other, wasting capital).
- EUR/USD + GBP/USD: Maximum combined exposure = 2.5% of account. If both are signaling the same direction, the lower-confidence signal is reduced by 50%.
- USD/CAD + XAU/USD: Both are commodity-correlated. If oil is spiking and gold is spiking simultaneously, total risk across these two pairs is capped at 1.5%.

News Event Blackout Schedule

The bot maintains an economic calendar integration (from Forex Factory API or DailyFX) to automatically identify and block trading around high-impact events:

- High Impact (Red): No new trades 30 minutes before and 15 minutes after the event. For XAU/USD: 45 minutes before, 20 minutes after.
- Medium Impact (Orange): Reduce position sizes by 50% for the affected pair.
- Low Impact (Yellow): No restriction.

6. TECHNICAL ARCHITECTURE

6.1 System Architecture Overview

APEX FX is built as a Python application that communicates with a MetaTrader 5 terminal via the official MetaTrader5 Python package. The architecture separates concerns into discrete, independently testable modules.

Layer	Components & Responsibilities
Data Layer	MT5 OHLCV feed, economic calendar feed, WTI crude oil feed (for USD/CAD), real-time tick subscription
Signal Engine	Per-instrument strategy modules (6 modules), Regime Detection Engine (RDE), indicator computation (TA-Lib / pandas-ta)
Risk Engine	Position sizing calculator, drawdown monitor, correlation controller, news event blackout manager
Execution Engine	Order router (MT5 API), spread checker, slippage monitor, retry logic with exponential backoff
State Manager	Open position tracker, daily P&L tracker, session state, regime state per pair
Monitoring Layer	Logging (structured JSON logs), Telegram bot alerts, performance dashboard (optional Grafana)
Config Layer	YAML-based per-instrument configuration files, environment variable secrets management

6.2 Technology Stack

Category	Technology	Purpose
Language	Python 3.11+	Core bot logic, strategy, risk
Trading Platform	MetaTrader 5	Order execution, data feed
MT5 Bridge	MetaTrader5 Python package (official)	MT5 API integration
Indicators	TA-Lib, pandas-ta	Technical indicator computation
Data Processing	pandas, NumPy	Time-series data handling
Economic Calendar	Forex Factory API / custom scraper	News event filtering
Secondary Data	Alpha Vantage or Quandl API	WTI crude oil price feed
Scheduling	APScheduler	Bar close event scheduling
Logging	Python logging + structlog	Structured JSON logs
Alerting	Telegram Bot API	Real-time trade alerts

Category	Technology	Purpose
Infrastructure	VPS (Windows Server, minimum 4GB RAM)	MT5 requires Windows
Version Control	Git + private repository	Code management
Testing	pytest + backtesting.py	Unit, integration, backtesting

6.3 Directory Structure

Project Structure

```
apex_fx_bot/ ├── main.py                    # Bot entry point ├── config/ │   ├── settings.yaml
# Global configuration │   ├── eurusd_config.yaml        # EUR/USD strategy params │   ├──
gbpusd_config.yaml        # GBP/USD strategy params │   ├── usdjpy_config.yaml        #
USD/JPY strategy params │   ├── usdchf_config.yaml        # USD/CHF strategy params │
├── usdcad_config.yaml        # USD/CAD strategy params │   └── xauusd_config.yaml        #
XAU/USD strategy params ├── strategies/ │   ├── base_strategy.py        # Abstract strategy
class │   ├── eurusd_strategy.py │   ├── gbpusd_strategy.py │   ├── usdjpy_strategy.py │   ├──
usdchf_strategy.py │   ├── usdcad_strategy.py │   └── xauusd_strategy.py ├── risk/ │   ├──
position_sizer.py        # ATR-based sizing │   ├── drawdown_monitor.py        # Daily/monthly
DD tracker │   ├── correlation_manager.py     # Cross-pair exposure control │   └──
news_filter.py            # Economic calendar integration ├── execution/ │   ├── order_router.py
# MT5 order placement │   ├── spread_checker.py        # Pre-trade spread validation │   └──
position_manager.py        # Open trade management ├── data/ │   ├── mt5_feed.py            #
OHLCV data provider │   ├── calendar_feed.py        # News calendar provider │   └──
commodity_feed.py        # WTI crude oil feed ├── regime/ │   └── regime_detector.py        #
RDE implementation ├── monitoring/ │   ├── logger.py              # Structured logging │   ├──
telegram_alerts.py        # Telegram notifications │   └── performance_tracker.py     # Trade stats
aggregation └── tests/   ├── test_strategies/   ├── test_risk/   └── test_execution/
```

6.4 Core Execution Loop

The bot's main loop runs on a scheduler that fires on every H1 bar close across all six instruments. The sequence per instrument per bar is:

1. Fetch latest OHLCV data (last 500 bars on H1 and H4, last 250 on D1)
2. Run RDE: classify current regime (TRENDING / RANGING / BREAKOUT / AVOID)
3. Run active strategy module: compute indicators, evaluate entry conditions
4. If signal exists: run pre-trade checks (spread, session gate, news blackout, daily DD, correlation limits)
5. If all checks pass: calculate position size via Position Sizer
6. Submit order via Order Router with hard stop-loss and take-profit
7. Log trade in structured format; send Telegram alert
8. Update State Manager: daily P&L, open positions, correlation exposure
9. Monitor existing open positions: check trailing stop activation conditions

7. BACKTESTING & VALIDATION FRAMEWORK

7.1 Backtesting Requirements

Before any strategy is promoted to live trading, it must pass a rigorous backtesting process. The backtesting framework must be walk-forward tested, not merely curve-fitted to historical data.

- Data quality: Tick-by-tick or M1 OHLCV data minimum. No synthetic data. Source from Dukascopy, HistData, or MT5 history.
- Test period: Minimum 5 years of historical data (2019–2024), covering at minimum: a bull trend, a bear trend, COVID-19 volatility spike (2020), and 2022 rate hike cycle.
- Transaction costs: Spreads, commissions, and slippage must be modeled. Use the broker's actual spread data; do not use fixed-spread assumptions.
- Walk-forward analysis: 70% in-sample for optimization, 30% out-of-sample for validation. Repeat rolling window across the test period.
- Monte Carlo simulation: Run 1,000+ Monte Carlo simulations to estimate the range of outcomes and maximum expected drawdown under random ordering of trades.
- Overfitting check: If a strategy requires more than 5 optimizable parameters to be profitable, it is considered overfit and rejected.

7.2 Minimum Backtest Acceptance Criteria

Metric	Minimum Acceptable	Reject If
Net Profit (5-year)	> 50%	< 20%
Profit Factor	> 1.3	< 1.1
Max Drawdown	< 20%	> 25%
Win Rate	> 45%	< 40%
Total Trades (5-year)	> 200	< 100
Out-of-Sample vs In-Sample Performance	> 70% of IS performance	< 50% of IS performance
Sharpe Ratio	> 0.8	< 0.5

7.3 Paper Trading Phase

Following successful backtesting, each strategy must complete a minimum 8-week paper trading (demo account) phase before live deployment. During this phase, all trades are logged, performance is tracked against backtest expectations, and slippage/spread assumptions are validated against real market conditions. Any strategy whose live paper-trading metrics deviate more than 30% from backtest metrics is returned for re-evaluation.

8. DEPLOYMENT & INFRASTRUCTURE

8.1 Infrastructure Requirements

Component	Specification
VPS OS	Windows Server 2019 or 2022 (required for MT5 terminal)
CPU	Minimum 4 vCPUs; 8 vCPUs recommended for real-time data processing
RAM	8 GB minimum; 16 GB recommended
Storage	100 GB SSD (MT5 history data + logs)
Network	< 10ms latency to broker server; 99.9% uptime SLA VPS
VPS Location	Collocate VPS in same data center as broker's trading server (NY4 Equinix for ECN brokers; London for European brokers)
Broker Account	ECN account with raw spreads + commission. Minimum $5,000 capital.
MT5 Terminal	Latest MT5 build; auto-update disabled during live trading hours

8.2 Monitoring & Alerting

- Telegram Bot: Real-time alerts for every trade open, close, stop-loss trigger, circuit breaker activation, and system error.
- Daily Report: At 22:00 UTC, the bot generates a daily performance summary (P&L, trades taken, win rate, drawdown) and sends via Telegram.
- Health Heartbeat: Bot sends a heartbeat ping every 5 minutes. If no heartbeat is received for 15 minutes, an emergency alert is sent and the broker's hosted EA (backup MT5 EA) closes all open positions.
- Error Logging: All exceptions are logged with full stack traces in structured JSON format. Logs rotate daily and are retained for 90 days.

8.3 Failover & Safety

- Primary: Python bot on VPS.
- Backup: A simple MT5 Expert Advisor (MQL5) that monitors for excessive drawdown and closes all positions if the Python bot becomes unresponsive. This EA is always running as a safety net.
- Broker SL: All orders are submitted with hard server-side stop-losses. The stop-loss is set at the broker level, not managed locally, so it persists even if the bot crashes or the VPS loses connectivity.
- Auto-Restart: Windows Task Scheduler restarts the Python bot automatically if it crashes.

9. DEVELOPMENT ROADMAP

Phase 1: Foundation (Weeks 1–4)

- Set up VPS, MT5 terminal, Python environment, broker ECN account
- Build data layer: MT5 feed, economic calendar feed, WTI crude feed
- Implement Regime Detection Engine (RDE) and validate against historical data
- Build risk engine: position sizer, drawdown monitor, news filter
- Unit tests for all risk modules

Phase 2: Strategy Development (Weeks 5–10)

- Develop and backtest EUR/USD strategy module
- Develop and backtest GBP/USD strategy module
- Develop and backtest USD/JPY strategy module (with BoJ intervention filter)
- Develop and backtest USD/CHF strategy module (with EUR/USD correlation filter)
- Develop and backtest USD/CAD strategy module (with WTI crude integration)
- Develop and backtest XAU/USD strategy module (with gold-specific news filter)

Phase 3: Integration & Paper Trading (Weeks 11–18)

- Integrate all 6 strategy modules with the execution engine and risk engine
- Build correlation manager to enforce cross-pair exposure limits
- Deploy to VPS and run 8 weeks of paper trading across all 6 pairs simultaneously
- Set up Telegram alerts, structured logging, daily performance reports
- Validate paper trading performance against backtesting benchmarks

Phase 4: Live Trading & Optimization (Weeks 19+)

- Deploy to live ECN account with minimum capital ($5,000 recommended)
- Start with 25% of target position sizes for first 4 weeks (reduced risk phase)
- Review weekly performance; adjust strategy parameters only during Friday close (not mid-week)
- Scale to full position sizes once 4 consecutive weeks of live trading meet backtest expectations
- Quarterly strategy review: full reoptimization and walk-forward revalidation

10. COMPLIANCE & RISK DISCLAIMER

APEX FX Trading Bot is intended for use by experienced traders who understand the risks of leveraged forex and commodity trading. Automated trading systems do not guarantee profits and can result in substantial losses.

Important Risk Disclosures

- Forex and gold trading carries a high risk of loss and may not be suitable for all investors.- Past backtesting performance does not guarantee future live trading results.- Leverage amplifies both gains and losses.- The bot should only be funded with capital that can be risked in its entirety.- This system does not constitute financial advice.- Users are responsible for ensuring their use of automated trading systems complies with their broker's terms of service and applicable local regulations.- The daily and account-level circuit breakers described in Section 5.5 are minimum protections, not guarantees against loss.

11. APPENDIX

A. Key Indicator Reference

Indicator	Formula / Params	Usage in APEX FX
ATR(14)	Average True Range, 14-period	Position sizing, stop-loss distance for all pairs
ADX(14)	Average Directional Index, 14-period	Regime detection: > 25 = Trending, < 20 = Ranging
EMA(20/50/200)	Exponential Moving Average	Trend direction bias, entry signal crossovers
RSI(14)	Relative Strength Index, 14-period	Entry confirmation, overbought/oversold filter
MACD(12,26,9)	Moving Avg Convergence Divergence	Trend momentum confirmation
Bollinger Bands(20,2)	20-period SMA ± 2 standard deviations	Mean reversion signals (USD/CHF), regime detection
Stochastic(5,3,3)	Stochastic oscillator	USD/JPY entry timing in oversold/overbought
Williams %R(14)	Williams Percent Range	USD/CHF mean reversion confirmation

B. Glossary

- ADR: Average Daily Range — the average pip movement of a pair over the last N trading days.
- ATR: Average True Range — a volatility measure that accounts for overnight gaps; the foundation of all stop-loss calculations in this system.

- Circuit Breaker: An automated rule that halts trading when a defined loss threshold is breached, protecting capital from catastrophic drawdown.
- ECN: Electronic Communication Network — a broker account type with raw spreads (no mark-up) plus a fixed commission per lot. Required for this system.
- Profit Factor: Total gross profit divided by total gross loss. A value > 1.0 means the system is profitable; > 1.5 indicates a robust strategy.
- RDE: Regime Detection Engine — the module that classifies market conditions as Trending, Ranging, or Breakout Pending to determine which strategy is active.
- Risk/Reward Ratio: The ratio of potential profit to potential loss on a single trade. A 1:2 RR means the target is twice the stop-loss distance.
- Sharpe Ratio: Risk-adjusted return measure. (Average Return - Risk-Free Rate) / Standard Deviation of Returns. Values > 1.0 are considered good.
- Slippage: The difference between the expected trade price and the actual execution price, caused by market movement between order submission and fill.
- Walk-Forward Analysis: A backtesting methodology that trains on one period and tests on a subsequent unseen period, rolling forward through time to prevent overfitting.

APEX FX TRADING BOT

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